

Linear Algebra: linear functionals

Chong Jongwon

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Contents

1	Dual Space	2
2	Double Dual	4
3	Transpose	5
4	Hyperspace	6
5	Annihilators	7
6	Transpose and Annihilators	8

1 Dual Space

Let V be a vector space over a field F .

Definition 1.1. A linear map $f : V \rightarrow F$ is called a *linear functional* on V .

Example. Let $F[x]$ be the set of polynomials over F . For each $\alpha \in \overline{F}$, define

$$L_\alpha : F[x] \rightarrow \overline{F} : p(x) \mapsto p(\alpha)$$

This is called the 'evaluation' at α . Clearly, it is a linear functional since

$$L_\alpha(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L_\alpha(p(x)) + L_\alpha(q(x))$$

Example. Let $C([a, b])$ be the space of continuous real-valued functions on $[a, b]$. Define

$$L : C([a, b]) \rightarrow \mathbb{R} : f \mapsto \int_a^b f(t) dt$$

L is clearly a linear functional by the linearity of the integral.

Definition 1.2. The collection of all linear functionals on V is called the *dual space* of V and is denoted as V^* .

Theorem 1.3. Let V be a finite-dimensional vector space over F , and let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V . Then, there exists a unique *dual basis* $\beta^* = \{f_1, \dots, f_n\}$ for V^* such that $f_i(\alpha_j) = \delta_{ij}$. Furthermore, for each linear functional $f \in V^*$,

$$f = \sum_{i=1}^n f(\alpha_i) f_i$$

and for each vector $\alpha \in V$,

$$\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$$

Proof. By the property of basis, for any $\alpha \in V$, there exist unique scalars $a_1, \dots, a_n \in F$ such that

$$\alpha = a_1 \alpha_1 + \dots + a_n \alpha_n$$

For each $1 \leq i \leq n$, define $f_i : V \rightarrow F$ by $f_i(\alpha) = a_i$ (or $f_i(\alpha) = ([\alpha]_\beta)_i$).

Then $f_i(\alpha_j) = \delta_{ij}$. Since $\dim V = \dim V^*$, it is enough to show that the set $\beta^* = \{f_1, \dots, f_n\}$ is linearly independent. Suppose that for some $c_i \in F$, $\sum_{i=1}^n c_i f_i = 0$. Then for each $1 \leq j \leq n$,

$$\left(\sum_{i=1}^n c_i f_i \right) (\alpha_j) = \sum_{i=1}^n c_i f_i(\alpha_j) = c_j = 0$$

Thus, β^* is linearly independent and forms a basis for V^* .

The second assertion can be proved as follows: if $\alpha \in V$ is represented as $\alpha = \sum_{i=1}^n x_i \alpha_i$ where $x_i \in F$, then for each $1 \leq j \leq n$:

$$f_j(\alpha) = f_j \left(\sum_{i=1}^n x_i \alpha_i \right) = \sum_{i=1}^n x_i f_j(\alpha_i) = x_j$$

Substituting x_j back into the representation of α , we obtain $\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$. □

Application: Lagrange Interpolation

Let $P_n(F) = \{p \in F[x] : \deg p \leq n\}$, and let $t_0, t_1, \dots, t_n \in F$ be distinct elements. For each $0 \leq i \leq n$, define the linear functional $L_i : P_n(F) \rightarrow F$ by $L_i(p) = p(t_i)$. These $\{L_0, L_1, \dots, L_n\}$ are linearly independent. Suppose that

$$c_0 L_0 + c_1 L_1 + \dots + c_n L_n = 0 \quad \dots (*)$$

where $c_i \in F$. By substituting the standard basis $\{1, x, x^2, \dots, x^n\}$ into $(*)$, we obtain a system of equations:

$$\begin{cases} c_0 + c_1 + \dots + c_n = 0 \\ c_0 t_0 + c_1 t_1 + \dots + c_n t_n = 0 \\ \vdots \\ c_0 t_0^n + c_1 t_1^n + \dots + c_n t_n^n = 0 \end{cases} \implies \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_0 & t_1 & \dots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_0^n & t_1^n & \dots & t_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix above is the **Vandermonde Matrix**, which is invertible since t_i are distinct. Thus $c_0 = c_1 = \dots = c_n = 0$. More precisely, define a linear map

$$T : P_n(F) \rightarrow F^{n+1} : p(x) \mapsto (p(t_0), \dots, p(t_n))$$

For $\beta = \{1, x, \dots, x^n\}$, $[T]_\beta$ is the **Vandermonde Matrix**.

If we choose a different basis $\beta' = \{1, (x - t_0), (x - t_0)(x - t_1), \dots, \prod_{j=0}^{n-1} (x - t_j)\}$, then $[T]_{\beta'}$ becomes:

$$\det[T]_{\beta'} = \det \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & t_1 - t_0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_n - t_0 & \dots & \prod_{j=0}^{n-1} (t_n - t_j) \end{bmatrix} = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0$$

Since the determinant is invariant with respects to the similar matrix, it is proved.

Theorem 1.4. $\{L_0, \dots, L_n\}$ is a dual basis for $P_n(F)^*$ with respect to the basis $\{P_0(x), \dots, P_n(x)\}$ for $P_n(F)$, where

$$P_i(x) = \prod_{j \neq i} \frac{x - t_j}{t_i - t_j}$$

Since $L_j(P_i) = P_i(t_j) = \delta_{ij}$, for any $p(x) \in P_n(F)$, we have:

$$p(x) = \sum_{i=0}^n L_i(p) P_i(x) = \sum_{i=0}^n p(t_i) P_i(x)$$

Furthermore, by the linearity of the integral, $\int_a^b p(x) dx = \sum_{i=0}^n p(t_i) \int_a^b P_i(x) dx$.

2 Double Dual

Question: Is every basis for V^ a dual of some basis for V ?*

Lemma 2.1. Let V be a finite dimensional vector space over F . For each $\alpha \in V$, define

$$\widehat{\alpha} : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, it is linear functional on V^* . Moreover, if $\widehat{\alpha}(f) = 0$ for any $f \in V^*$, then $\alpha = 0$.

Note: the last assertion can be said: $(V^*)^0 = \{0\}$.

Proof. Let $f, g \in V^*$ and $c \in F$ be given. Then,

$$\widehat{\alpha}(cf + g) = (cf + g)(\alpha) = cf(\alpha) + g(\alpha) = c\widehat{\alpha}(f) + \widehat{\alpha}(g)$$

Thus proved. To show the second assertion, let $\alpha \neq 0$. Then, there exists a basis β for V containing α . Then, there is a dual basis β^* for V^* . Take $f_\alpha \in \beta^*$ corresponding to α . Then, $f_\alpha(\alpha) = 1 \neq 0$. Thus proved. $\square$

Theorem 2.2. Let V be a finite dimensional vector space over F . Then, the map

$$V \rightarrow V^{**} : \alpha \mapsto \widehat{\alpha}$$

is an Isomorphism.

Proof. Let $\alpha, \beta \in V$ and $c \in F$, and for any $f \in V^*$,

$$(\widehat{c\alpha + \beta})(f) = f(c\alpha + \beta) = cf(\alpha) + f(\beta) = c\widehat{\alpha}(f) + \widehat{\beta}(f) = (c\widehat{\alpha} + \widehat{\beta})(f) = \widehat{c\alpha + \beta}(f)$$

Moreover, by the Lemma, the kernel of the map is trivial. Thus injective, and surjective because:

$$\dim V^{**} = \dim V^* = \dim V$$

From the Theorem above, we obtain that:

If $\widehat{\alpha} \in V^{**}$, then there exists a unique $\alpha \in V$ such that for all $f \in V^*$, $\widehat{\alpha}(f) = f(\alpha)$.

We can restate this result as:

Corollary 2.3. Let V be a finite dimensional vector space over F .

Every basis for V^* is the dual of some basis for V .

Proof. Let $\{f_1, \dots, f_n\}$ be a basis for V^* . Then, there is a dual basis $\{\widehat{\alpha}_1, \dots, \widehat{\alpha}_n\}$ for V^{**} such that $\widehat{\alpha}_i(f_j) = \delta_{ij}$. By the above corollary, for each $\widehat{\alpha}_i \in V^{**}$, there exists a unique $\alpha_i \in V$ such that $\widehat{\alpha}_i(f) = f(\alpha_i)$ for all $f \in V^*$. Now, $\widehat{\alpha}_i(f_j) = f_j(\alpha_i) = \delta_{ji}$, the claim is proved. $\square$

3 Transpose

Theorem 3.1. Let V and W be a vector space over F , and let $T : V \rightarrow W$ be a linear map. Then,

$$T^t : W^* \rightarrow V^* : g \mapsto g \circ T$$

is a linear map.

Furthermore, if V and W have finite dimensional with ordered basis β and γ for V and W , respectively, then

$$([T]_{\beta}^{\gamma})^t = [T^t]_{\gamma^*}^{\beta^*}$$

Proof. Since the composition of two linear maps is linear, the map T^t is well-defined. Given $f, g \in W^*$ and $c \in F$,

$$T^t(cf + g) = (cf + g) \circ T = cf \circ T + g \circ T = cT^t(f) + T^t(g)$$

Thus, the first assertion is proved.

Now, suppose that V and W have finite dimensional. Let $\beta = \{x_1, \dots, x_n\}$ and $\gamma = \{y_1, \dots, y_m\}$.

Then, there are dual basis for V^* , W^*

$$\beta^* = \{f_1, \dots, f_n\}, \quad \gamma^* = \{g_1, \dots, g_m\}$$

respectively. Observe that: for each $1 \leq j \leq m$,

$$T^t(g_j) = g_j \circ T = \sum_{s=1}^n (g_j \circ T)(x_s) f_s$$

Meanwhile, denote $A = [T]_{\beta}^{\gamma}$. Then,

$$(g_j \circ T)(x_i) = g_j(T(x_i)) = g_j\left(\sum_{k=1}^m A_{ki} y_k\right) = \sum_{k=1}^m A_{ki} g_j(y_k) = \sum_{k=1}^m A_{ki} \delta_{jk} = A_{ji}$$

Therefore,

$$[T^t]_{\gamma^*}^{\beta^*} = [[T^t(g_1)]_{\beta^*} \quad \dots \quad [T^t(g_m)]_{\beta^*}] = A^t = ([T]_{\beta}^{\gamma})^t$$

□

Theorem 3.2. Let F be a field, and let $T : F^n \rightarrow F^m$ be a map. A map T is linear map if and only if:

There exist $T_1, \dots, T_m \in (F^n)^*$ such that for all $x \in F^n$, $T(x) = (T_1(x), \dots, T_m(x))$

Proof. Let $\{g_1, \dots, g_m\}$ be a dual basis for F^m with respects to standard basis for F^m . For each $1 \leq i \leq m$, define

$$T_i = T^t(g_i)$$

Let $x \in V$ be given. Then, for each $1 \leq i \leq m$,

$$T_i(x) = T^t(g_i)(x) = (g_i \circ T)(x) = g_i(T(x)) = ([T(x)]_{\gamma})_i$$

Thus proved. The converse is trivial.

□

4 Hyperspace

Lemma 4.1. Let $f, g : V \rightarrow F$ be linear functionals on V .

$$g = c \cdot f \text{ for some } c \in F \text{ if and only if } \ker f \leq \ker g.$$

Proof. If $f = 0$, then it is trivial. Assume $f \neq 0$. Then $\ker f$ is a hyperspace of V .

Let $\alpha \in V$ such that $f(\alpha) \neq 0$. Put $c = g(\alpha)/f(\alpha)$.

Then, $g - cf$ is a linear functional such that $\ker f \leq \ker(g - cf)$ and $\alpha \in \ker(g - cf)$. Thus $\ker(g - cf) = V$, or $g - cf = 0$. $\square$

Theorem 4.2. Let $g, f_1, \dots, f_r$ be linear functionals on V . Then,

$$g \text{ is a linear combination of } \{f_1, \dots, f_r\} \text{ if and only if } \bigcap_{i=1}^r \ker f_i \leq \ker g.$$

Proof. If $g = \sum_{i=1}^r c_i f_i$ for some $c_i \in F$, then for any $\alpha \in \bigcap_{i=1}^r \ker f_i$, $g(\alpha) = \sum_{i=1}^r c_i f_i(\alpha) = 0$.

Conversely, we shall use the mathematical induction for $r \in \mathbb{N}$.

By the previous Lemma, if $r = 1$, it is proved. Suppose that it is held for $r = k - 1$.

Let $f_1, \dots, f_k$ be linear functionals on V such that $\bigcap_{i=1}^k \ker f_i \leq \ker g$.

Let $g, f_1, \dots, f_{k-1}$ be restricted by $\ker f_k$. That is,

$$f_i|_{\ker f_k} : \ker f_k \rightarrow F : \alpha \mapsto f_i(\alpha) \quad (i = 0, \dots, k-1, \text{ where } f_0 = g)$$

And, let $\alpha \in \ker f_k$ such that for all $1 \leq i \leq k-1$, $f_i(\alpha) = 0$.

That is, $\alpha \in \bigcap_{i=1}^k \ker f_i$ and therefore $g|_{\ker f_k}(\alpha) = 0$ by the hypothesis. By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i f_i|_{\ker f_k} \text{ for some } c_i \in F.$$

Put $h = g - \sum_{i=1}^{k-1} c_i f_i$. Now, for any $\alpha \in \ker f_k$, $h(\alpha) = 0$, or $\ker f_k \leq \ker h$.

The previous Lemma gives that $h = c_k f_k$ for some $c_k \in F$. Now, $g = \sum_{i=1}^k c_i f_i$. $\square$

Theorem 4.3. Let V be a vector space over F .

For any non-zero linear functional $f \in V^*$, $\ker f$ is a hyperspace of V .

Conversely, every hyperspace in V is the kernel for some linear functional on V .

Proof. Let $f \in V^*$ be a non-zero linear functional. Since f is non-zero, choose $\alpha \in V \setminus \ker f$ (or $f(\alpha) \neq 0$).

We shall show that $V = \ker f + \langle \alpha \rangle$. Let $\beta \in V$ be given. Put $c = f(\beta)/f(\alpha)$. (It can be defined because $f(\alpha) \neq 0$).

Then, $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$, thus $\beta - c\alpha \in \ker f$, or $\beta \in \ker f + \langle \alpha \rangle$.

Conversely, let $N \leq V$ be a hyperspace. Let $\alpha \in V \setminus N$. Since N is a hyperspace, $N + \langle \alpha \rangle = V$. Now, for each $\beta \in V$,

$$\beta = \gamma + c\alpha \text{ for some } \gamma \in N, c \in F.$$

This γ and c are uniquely determined because $N \cap \langle \alpha \rangle = \{0\}$. Define $g : V \rightarrow F$ as $g(\beta) = c$. Then, $\ker g = N$. $\square$

5 Annihilators

Definition 5.1. Let V be a vector space over F , and let $S \subseteq V$ be a subset. The annihilator of S is the set

$$S^0 \stackrel{\text{def}}{=} \{f \in V^* \mid f(\alpha) = 0 \text{ for all } \alpha \in S\}.$$

Note: $f \in S^0 \iff S \subseteq \ker f$.

Theorem 5.2. Let V be a finite-dimensional vector space, and let $W \leq V$ be a subspace. Then

$$\dim W + \dim W^0 = \dim V.$$

Proof. Clearly, $W^0 \leq V^*$. Let $\{\alpha_1, \dots, \alpha_k\}$ be a basis for W .

Choose $\alpha_{k+1}, \dots, \alpha_n \in V$ such that $\{\alpha_1, \dots, \alpha_n\}$ is a basis for V .

Let $\{f_1, \dots, f_n\}$ be the dual basis for V^* corresponding to $\{\alpha_1, \dots, \alpha_n\}$.

For $k+1 \leq i \leq n$, we have $f_i(\alpha_j) = 0$ for all $1 \leq j \leq k$. Thus, $f_i(\alpha) = 0$ for all $\alpha \in W$, which implies $\{f_{k+1}, \dots, f_n\} \subseteq W^0$. Clearly, this set is linearly independent. To show that it spans W^0 , let $f \in W^0 \leq V^*$ be given. Then

$$f = \sum_{i=1}^n f(\alpha_i) f_i = \sum_{i=k+1}^n f(\alpha_i) f_i \in \text{span}\{f_{k+1}, \dots, f_n\}$$

since $f(\alpha_i) = 0$ for $1 \leq i \leq k$. Thus, $\dim W^0 = n - k = \dim V - \dim W$. □

Corollary 5.3. If $W \leq V$ with $\dim W = k < n = \dim V$, then

$$W = \ker f_{k+1} \cap \dots \cap \ker f_n$$

where f_i are the functionals described in the theorem above.

Corollary 5.4. For subspaces $W_1, W_2 \leq V$ with $\dim V < \infty$,

$$W_1 = W_2 \iff W_1^0 = W_2^0.$$

Proposition 5.5. Let V be a vector space over F , and let $W_1, W_2 \leq V$ be subspaces. Then:

1. $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$
2. $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$

6 Transpose and Annihilators

Theorem 6.1. Let V and W be vector spaces over F , and let $T: V \rightarrow W$ be a linear map. Then

$$\ker T^t = (\operatorname{Im} T)^0.$$

In particular, if V and W are finite-dimensional, then:

1. $\operatorname{rank} T^t = \operatorname{rank} T$
2. $\operatorname{Im} T^t = (\ker T)^0$

Proof. The first assertion is proved as follows:

$$g \in \ker T^t \iff T^t(g) = 0 \iff g(T(\alpha)) = 0 \text{ for all } \alpha \in V \iff g(\gamma) = 0 \text{ for all } \gamma \in \operatorname{Im} T \iff g \in (\operatorname{Im} T)^0.$$

Thus, $\ker T^t = (\operatorname{Im} T)^0$. Now, suppose that V and W have finite-dimensional.

Put $n = \dim V$ and $m = \dim W$ and $r = \operatorname{rank} T$. Since $\dim \operatorname{Im} T + \dim (\operatorname{Im} T)^0 = \dim W$, $\dim (\operatorname{Im} T)^0 = m - r$. Since the first result of this theorem with the Dimension Theorem,

$$m - r = \dim \ker T^t = \dim W^* - \dim \operatorname{Im} T^t = m - \operatorname{rank} T^t$$

Therefore, $\operatorname{rank} T^t = m$. To show the last assertion, let $f \in \operatorname{Im} T^t$. That is, $f = T^t(g)$ for some $g \in W^*$. Now,

$$f(\alpha) = T^t(g)(\alpha) = (g \circ T)(\alpha) = g(0) = 0 \text{ for all } \alpha \in \ker T$$

Thus $f \in (\ker T)^0$, or $\operatorname{Im} T^t \subseteq (\ker T)^0$. Finally,

$$\dim (\ker T)^0 = \dim V^* - \dim \ker T = \operatorname{rank} T = \operatorname{rank} T^t$$

Proved. □